RuFaS Model Review for Version 1.0 Release

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1 Introduction

1.1 Objective

The objective of the evaluation before releasing v1.0 was to assess integrated, system-level behavior of the Ruminant Farm Systems (RuFaS) model (`https://www.rufas.org/`) following implementation of methodological updates. While individual modules and pull requests undergo routine testing and review during development, this evaluation focused specifically on whole-farm performance, confirmation of expected changes and similarities with previous version release, and alignment with expected biological and environmental responses.

1.2 Key Findings

Milk Production Closely Aligned with Farm-Reported Values: Across the 13 evaluation farms, predicted annual milk production closely matched reported herd-level production, with all farms falling within $\pm 2\%$ of reported values. Deviations were attributable to identifiable input alignment issues (e.g., parity structure and milking fraction definitions) rather than structural model changes. These results indicate that steady-state herd dynamics and lactation curve parameter calibration behave consistently across diverse herd sizes and management systems.

Environmental Intensities Were Consistent with Published Literature Benchmarks: Estimated emission intensities for enteric methane, manure methane, manure nitrous oxide (N_2O), and feed-related emissions were within ranges reported in peer-reviewed dairy life cycle assessment (LCA) and emissions studies.

- For enteric methane, both intensity-based metrics (e.g., per unit of milk) and absolute annual per-cow emissions for 11 and 12 farms, respectively, were within literature-derived plausibility bounds, indicating no systematic over- or under-estimation relative to published benchmarks.
- Manure methane and manure N_2O estimates varied in expected directions based on manure management system type and regional climate conditions, consistent with established biophysical relationships reported in the literature.
- Feed-related emission factors, with and without land use change assumptions, produced carbon intensity estimates comparable to those reported in national- and sector-level assessments.

Cases (3) in which estimates fell outside typical literature ranges were attributable to identifiable farm-specific drivers—such as production efficiency (e.g., milk-to-feed ratios) or distinct manure handling configurations—rather than to model instability or unexplained variability. This pattern suggests that observed variation reflects sensitivity to biologically and management-relevant factors.

Changes in Model Outputs From Version 0.8 to 1.0 Were Consistent with Documented Methodological Updates: Differences between outputs in v0.8 and v1.0 reflect intentional and documented methodological updates rather than unexplained variability. Specifically, changes in results were driven by:

- Refinements to dietary net energy supply calculations and algorithm underlying dry matter intake predictions,
- Structural revisions to the manure module including: implementation of a time-lagged sinusoidal curve to model liquid manure temperature, revised heifer manure methane potential for solid manure systems, emission factor responsiveness to compost management, and removal of bedding-derived volatile solids from methane calculations in solid manure systems.

Consistent with these updates, modeled methane and N_2O yields changed in expected directions. For example:

- Liquid manure systems showed climate-dependent changes in methane emissions, reflecting the revised temperature-driven calculations introduced in the manure module.
- Solid manure storage systems showed reduced methane emissions, consistent with a lower methane potential of heifer manure and removal of bedding volatile solids from methane estimation.

Taken together, these results indicate that v1.0 updates produced coherent, mechanism-driven changes in emission estimates at the whole-farm scale. The alignment between model behavior and the underlying methodological revisions supports the interpretability and internal consistency of the updated model

Internal System Coherence Was Maintained: Production, intake, emissions, and carbon intensity metrics remained internally coherent across modules. For example, changes in feed efficiency (milk / feed intake) were reflected in subsequent enteric, manure, and feed emissions intensities. Changes in milk yield and dry matter intake were reflected in emission intensities, and emission yields per unit of intake behaved consistently with updated methods. No evidence was observed of unintended cross-module instability or non-biological system behavior, for example, emissions decreasing with increased feed intake under otherwise comparable conditions. These results indicate that the modules remained well-integrated and that system-wide outputs were predictable and interpretable after the updates.

1.3 Evaluation Methodology

1.3.1 Farm Selection and Simulation Design

A subset of 13 farms was selected from the pilot testing dataset to represent variation in:

- Herd size
- Milk production level
- Geographic location
- Manure management systems
- Feeding and management practices

Each farm configuration was simulated four independent times to account for stochastic variation within the model. Production and environmental outputs were averaged across runs prior to evaluation to reduce noise attributable to stochastic processes. The results reported here are based on use of the NASEM nutrient requirement model (NASEM, 2021), the Niu, Kebreab, Hristov, et al. (2018) lactating cow enteric methane model, the Mills et al. (2003) enteric methane model for dry cows, and the IPCC (2006) enteric methane model for youngstock.

1.3.2 Outputs Evaluated

The evaluation included both production and environmental metrics spanning the full farm system:

- **Production and herd metrics:**
  - Milk yield and Fat and Protein Corrected Milk (FPCM;International Dairy Federation, 2022) yield (kg/cow/day; annual herd production)
  - Dry matter intake (lactating, non-lactating, and herd-level)
  - Feed efficiency metrics (milk production/ whole herd or lactating cow intake)
  - Herd demographic structure (parity distribution, turnover, age at first calving, calving interval)
  - Metabolic body weight and allocation factors
- **Environmental Metrics:**
  - Enteric methane emissions (production per cow per day and year; intensity [kg CH_4/ kg FPCM])
  - Manure methane emissions (per-cow production; yield per manure volatile solids; intensity [kg CH_4/ kg FPCM])
  - Manure N_2O emissions (per-cow production, yield per manure N; intensity [kg CH_4/ kg FPCM])
  - Feed-related emissions (yield per kg intake and intensity [kg CO_2-eq/kg FPCM] with and without land use change)
  - Carbon intensity of milk (CO_2-eq/kg FPCM)

1.3.3 Evaluation Criteria

Model performance was assessed using three complementary approaches:

Comparison with Farm-Reported Production: Predicted annual herd milk production was compared against reported farm-level production. Percent difference and absolute deviation were calculated to assess calibration of steady-state herd dynamics and production outputs.

Benchmarking Against Literature Ranges: Environmental intensities and yields were compared against ranges reported in peer-reviewed dairy life cycle assessment (LCA) and emissions studies literature. Literature ranges were used as plausibility bounds rather than strict validation thresholds. Instances falling outside expected ranges were examined in relation to production characteristics (e.g., milk:feed ratios, manure system configuration) to determine if extreme values aligned with expected behavior of the modeled system.

Version to Version Comparison: Outputs from v0.8 and v1.0 were compared for the same farm configurations. Differences were interpreted in the context of documented methodological changes, including:

- Updates to herd demographics simulation
- Lactation curve implementation
- Diet formulation refinements
- Structural revisions to the manure module (liquid and solid storage temperature modeling, methane potential updates, e mission factor responsiveness, and removal of bedding volatile solids from methane estimation)

Because updates affected milk production and feed dry matter intake predictions, emission comparisons were interpreted on an intake (kg/kg DMI) basis where appropriate to isolate methodological effects from production-driven intensity changes.

1.3.4 Interpretation Framework

This evaluation was designed as a structured assessment of model plausibility and integrated system performance rather than a formal statistical validation or calibration study. No parameter re-fitting was conducted during the evaluation process. Model outputs were reviewed to assess whether system behavior was reasonable, interpretable, and aligned with expectations. Specifically, outputs were examined for:

- Biological realism, defined as consistency with known physiological and biogeochemical processes;
- Consistency with management inputs, such that changes in feeding, housing, or manure handling were reflected in outputs in expected ways;
- Internal coherence across modules, including aligned responses in intake, production, emissions, and intensity metrics;
- Directional alignment with documented methodological updates, ensuring that version-to-version changes reflected intended model revisions; and
- Position within reasonable literature-reported ranges, recognizing expected variability across systems and studies.

1.4 Scope and Limitations

The evaluation sample consisted of 13 farms representing a range of diverse dairy systems but does not constitute a statistically representative national dataset. While this diversity supports assessment of model behavior across contrasting scenarios, the sample is not intended to support population-level inference or national benchmarking. Comparisons of model outputs to published literature rely on published results that vary in methodology, assumptions and system boundary definitions. Therefore, results should be interpreted as evidence that the updated model exhibits reasonable, stable, and biologically plausible behavior under diverse scenarios rather than direct validation against a population-wide representative empirical standard. Results from this evaluation do not constitute definitive empirical validation or quantification of model accuracy relative to observed data.

2 Evaluation Results

2.1 Carbon Footprint of Milk Version 1.0

Table 1:

Variable	Average	Median	Min	Max
Milk CO2 Intensity w/o LUC	1.211	1.211	0.799	1.777
Enteric Intensity	0.516	0.493	0.410	0.676
Manure Intensity	0.288	0.268	0.042	0.608
Feed Intensity w/o LUC	0.407	0.372	0.150	0.634
Milk CO2 Intensity w/ LUC	1.405	1.399	0.945	2.220
Feed Intensity w LUC	0.601	0.618	0.281	1.032
Feed LUC Intensity	0.194	0.120	0.016	0.585

Table 2: Relative contribution of each emission source to the total carbon footprint of milk with and without accounting for land use change emissions.

Without LUC	**Average**	**Median**	**Min**	**Max**
Enteric Footprint Contribution	43.8%	42.8%	32.7%	55.3%
Manure Footprint Contribution	23.2%	24.1%	3.8%	34.2%
Feed Footprint Contribution	33.0%	32.1%	18.8%	54.2%
With LUC				
Enteric Footprint Contribution	38.0%	38.0%	26.2%	49.4%
Manure Footprint Contribution	19.8%	20.3%	3.7%	27.4%
Feed Footprint Contribution	29.5%	31.1%	10.8%	52.6%
LUC Footprint Contribution	12.7%	8.5%	1.2%	42.3%

- RuFaS predicted carbon intensities cover a wide range that reflect the diversity of the simulated farms
- The average and range of intensities align with other national assessments for carbon intensity (e.g. Pelton et al., 2025, Rotz et al., 2024, Capper and Cady, 2020)

2.2 Production Outcomes

2.2.1 Milk Production & DMI Outcomes

Table 3: Select production outcome summary statistics from the 13 evaluation farms

Production Metrics	**Units**	**Avg**	**Median**	**Min**	**Max**
Milk Production	kg/cow/d	35.0	34.9	22.8	43.1
FPCM Production	kg/cow/d	34.7	34.0	21.7	43.3
Lac Cow DMI	kg/cow/d	25.7	25.3	19.9	31.7
Growing Heifer DMI	kg/cow/d	9.7	10.1	7.4	11.4
Close-Up DMI	kg/cow/d	15.3	16.3	11.0	20.5
Herd Feed Efficiency	kg milk/ kg herd DMI	0.91	0.95	0.63	1.06
Lac Cow Feed Efficiency	kg milk/kg cow DMI	1.36	1.33	1.09	1.66
Lac Cow FPCM Feed efficiency	kg FPCM/kg cow DMI	1.35	1.34	1.09	1.66
Mature Metabolic Bodyweight	kg	690	681	571	747
Milk Allocation Factor		0.89	0.89	0.84	0.94

2.2.2 Herd Demographics

Table 4: Herd demographic summary statistics from the 13 evaluation farms

Demographic Metrics		**Average**	**Median**	**Min**	**Max**
Cows in Parity 1	%	36.24%	35.59%	27.03%	45.88%
Cows in Parity 2	%	25.26%	25.36%	21.87%	29.52%
Cows in Parity 3	%	16.04%	16.44%	12.06%	19.77%
Cows in Parity 4	%	10.35%	10.68%	6.72%	13.50%
Youngstock % of herd (excluding springers)	%	47.71%	48.06%	28.78%	63.91%
Herd Turnover	%	37.69%	37.67%	27.04%	47.32%
Age at First Calving	d	695	689	665	757
Calving Interval	d	396	392	376	447

2.3 Environmental Benchmarking

2.3.1 Enteric Methane

Table 5: Enteric methane summary statistics from the 13 evaluation farms

Enteric Methane	**Units**	**Average**	**Median**	**Min**	**Max**
Lac Cow Production					
RuFaS predictions	kg CH4/ cow /d	0.4517	0.4405	0.3940	0.5524
Lit. Range				0.0790	0.7290
RuFaS predictions	kg CH4 / lact cow / yr	165.1902	160.7975	143.8020	201.6085
Lit. Range				128	188
Enteric Methane Intensity					
RuFaS predictions	whole herd kg CH4 / kg FPCM	0.0211	0.0196	0.0160	0.0297
Lit. Range				0.011	0.023

Of the 13 farms, 3 were above the target range from literature benchmarks for enteric methane emissions intensity (e.g. Niu, Kebreab, Hristov, et al., 2018) though still within the reported ranges

for emissions per cow per day. These farms correspond to lower production systems with lactating cow feed efficiency below 1.3 kg milk / kg DMI.

2.3.2 Manure Methane

Table 6: Manure Methane summary statistics from the 13 evaluation farms

Manure Methane	Units	Average	Median	Min	Max
Herd Intensity	whole herd kg CH4	0.0074	0.0062	0.0006	0.0243
Lit. Range	/ kg FPCM			0.001	0.020
Production Per Cow	kg CH4 / cow / yr	81.6	73.6	4.0	242.5
Lit. Range				4.7	375.2
Yield	kg CH4/kg VS	0.0203	0.0190	0.0013	0.0566
Lit. Range				0.001	0.023

One farm was above the target range for manure methane intensity. This farm was also one of the farms that was above the target range for enteric methane intensity and can also be attributed to the low milk:feed ratio.
One farm was below the expected minimum range for manure methane intensity and annual manure methane production per cow. This farm uses predominantly daily spread for management of its lactating cow manure and has very little long term manure storage leading to low manure methane emmission.

2.3.3 Manure N2O

Table 7: Manure Nitrous Oxide summary statistics from the 13 evaluation farms

Manure Nitrous Oxide	Units	Average	Median	Min	Max
Herd Intensity	whole herd kg N2O	1.80E-04	1.23E-04	0.00	6.68E-04
Lit. Range	/ kg FPCM			0.00009	0.00033
Manure N2O Production Per Cow	kg N2O / lact cow / yr	1.79	1.41	0.00	5.89
Lit. Range				0	4.5
Manure N2O Yield	kg N2O /kg Manure N	0.0087	0.0068	0.0000	0.0275
Lit. Range				0.0001	0.06

All farms were within the range of published literature results for N_2O emissions (e.g. Owen and Silver, 2015), except for Farms 5 & 9 which was above the reported range of N_2O emissions per kg FPCM. These relatively high N_2O emissions are driven by the use of compost-bedded pack barns for the growing and close-up animals which has an N_2O emission yield of 0.07 kg N_2O/kg N excreted.

2.3.4 Feed Emissions

Table 8: Diet emission carbon yield per kg dry matter of feed summary statistics

Feed Carbon Yield	Units	Average	Median	Min	Max
Feed CO2-eq w/o LUC	Whole herd	0.3449	0.3662	0.0938	0.5762
Feed LUC CO2-eq	kg CO2-eq	0.1832	0.1040	0.0137	0.6741
Total Feed w/ LUC	/kg DMI	0.6039	0.6272	0.2992	0.9016

2.4 Alignment with Farm Reported Production

Table 9: Summary of RuFaS predicted milk production and alignment with reported herd annual milk production[1]

Farm Reported Milk	Units	Average	Median	Min	Max
Herd Reported Annual Milk	tons	30,358	24,021	427	103,138
Herd Size	cow	2,689	1,935	51	9,333
RuFaS Predicted Average Milk Per Cow	kg/cow	35.0	34.9	22.8	43.1
RuFaS Predicted Annual Milk as % of Farm Reported Milk	%	99.4%	99.4%	99.34%	100.68%
Predicted Annual Herd Milk % Diff. from Reported	%	0.77%	-0.76%	-1.66%	0.68%

[1]Metrics for comparison with farm reported milk production is based on 12 farms as one farm did not provide their annual milk production

Notes on Comparison with Farm Reported Milk Production

- Predicted milk was within 2% of farm provided annual milk production
- Further user guidance is needed to advise users on importance of parity structure inputs and alignment with RuFaS predicted steady state herd to ensure predicted milk matches target milk
- Further refinement in recommendations for model use to increase precision and accuracy of RuFaS predicted milk is needed. Including:
  - Variation in predicted milk for different herd sizes
  - Identification of best practices for herd demographic and herd management input definition to match target milk and herd structure

2.5 Comparison with Previous Results

2.5.1 Change in Predicted Milk from v0.8 to v1.0

Notes on Change in Predicted Milk from v0.8 to v1.0

- The changes in milk yield between the evaluations of v0.8 and v1.0 can be attributed to input changes made via trial-and-error during the v0.8 pilot testing process that caused the final

Table 10: Summary of the Change in Production Outcomes between v0.8 and v1.0 Scenarios.

Production Differences from 2024	Units	Absolute Average	Median	Min	Max
Milk per Cow	kg/cow/d	-0.51	-0.429	-1.47	-0.068
DMI per Lac Cow	kg/cow/d	1.63	2.30	-3.05	5.33
Herd Intake (kg/yr)	%	3.8%	4.0%	-6.4%	14.5%
Lact Cow Intake (kg/herd/yr)	%	7.8%	8.4%	-7.8%	23.7%
Non-Lac Cow intake (kg/herd/yr)	%	-1.8%	-3.6%	-15.0%	23.9%

inputs to differ from farm-reported values. These deviations were not replicated in the v1.0 evaluation process in preference for adherence to farm-reported values.

- The differences in milk production combine with updates to diet formulation methods to result in some change in predicted DMI as well as total milk production.
- Due to these changes in production, environmental outcomes between v0.8 and v1.0 are more appropriately compared per unit of DMI, not per unit of milk production.

2.5.2 Predicted Emmision Yield Change from v0.8 to v1.0

Table 11: Summary of the Change in enteric and manure emission yield (kg gas/kg DMI)

Emmision Yield Percent Change	Absolute Average	Median	Min	Max
Herd Enteric Emission Yield (kg CH4/ kg DMI)	3.79%	0.01	-5.37%	9.58%
Lact Cow Enteric Emission Yield (kg CH4/kg Lact cow DMI)	3.67%	-0.50%	-6.41%	12.25%
Herd Manure CH4 Yield	51.89%	-52.42%	-90.01%	37.90%
Herd Manure N2O Yield	46.34%	-35.09%	-100.00%	0.56%

- Two farms (Farm 7 and Farm 10) saw a greater than 5% increase in Enteric Emission yield which can be attributed to a change in the lactating cow enteric emission yield.
- The changes in manure emissions are attributed to fundamental changes in the Manure Module methodology.

2.5.3 Updates to Manure Module Liquid Storage Methodology

- Structural changes in how the Manure Module represents manure entering and leaving each manure management process allowed for sequential modeling of manure storage and processing. There is no systematic effect of these structural changes on total emissions.
- Changes to Manure Temperature Modeling in Liquid Manure Systems caused the largest change in emissions when compared to the v0.8 values.
  - The effect of this change for farms with liquid manure storage is increased manure methane in warm climates and decreased manure methane in cold climates
- A minimum temperature and emptying amount was implemented for manure lagoon storage
  - The effect will be a variable increase in emissions depending on the climate and emptying timing

Impact of Manure Module Updates on Liquid Manure Systems

Case Farm 1: Warm Climate Liquid Storage

Farm Characteristics:

- 3,500 Cow Herd in California
- All Freestall housing
- All pens flushed to rotary screen separator
- Anaerobic digestion followed by storage in anaerobic lagoon
- Average Annual Temp: $\sim$ 18°C (64°F)
- Max Daily Avg Temp: $\sim$ 36°C (97°F)

The resulting Manure Methane Yield is: **0.0412** kg CH_4 per kg Manure VS, which is in line with the yield per kg manure VS predicted by USDA GHG Entity scale prediction methods (Hanson, Itle, and Edquist, 2024).

Take Home: Manure Methane emissions increased as expected.
Reason: Improvements in manure temperature modeling (PR #2636) and representation of anaerobic lagoons (PR# 2638) resulted in increased methane emissions from the anaerobic lagoon on this warm climate farm.

Case Farm 2: Cool Climate Liquid Storage

Farm Characteristics:

- 1,950 Cow Herd in New York
- All Freestall housing
- Sand and manure solids bedding
- Pens scraped manually to rotary screen
- Anaerobic Lagoon (12 mo) and slurry storage (6 mo)
- Average Annual Temp: $\sim$ 9.5°C (49°F)
- Max Daily Average Temp: $\sim$ 28°C (82°F)

The resulting Manure Methane Yield is: **0.0185** kg CH_4 per kg Manure VS, which is in line with the yield per kg manure VS predicted by USDA GHG Entity scale prediction methods (Hanson, Itle, and Edquist, 2024) .

Compared to RuFaS v0.8 predictions, the updated method results in a **58% decrease** in the manure methane emission yield in units of kg CH_4 per kg DMI.

2.5.4 Updates to Manure Module Solid Storage Methodology

- Similar to liwuid systems, structural changes in how the Manure Module represents manure entering and leaving each manure management process allowed for sequential modeling of manure storage and processing. There is no systematic effect of these structural changes on total emissions.
- The methane potential of manure from youngstock was updated to reflect the lower methane potential of manure from this animal class.
  - Farms with solid storage will see a decrease in emissions from this change
- The bedding volatile solids was removed from the estimation of solid storage methane production.
  - Farms with solid storage will see a decrease in emissions from this change
- Composting and Compost Bedded Pack Barn emissions factors were updated to be more responsive to management changes (i.e. turning + mixing frequency).
  - Farms with intensive management of these systems will see a decrease in their emissions

Case Farm 3: Cool Climate Solid Storage

Farm Characteristics:

- 58 Cow Farm in Wisconsin
- Tie stalls & daily spread for lac. cows
- Bedded-pack barns for heifers and dry cows
- Mixed bedded-pack emptied every 3 mos
- Average Annual Temp: 9°C (48°F)
- Max Daily Average Temp: 29°C (84°F)

The whole farm Manure Methane Yield is: **0.0012** kg CH_4 per kg manure VS. The Manure Methane Yield for heifer and dry cow bedded pack barns are 0.0017 and 0.0006 kg CH_4 per kg manure VS, respectively. These precisely match expectations from USDA Entity Scale emissions methods (Hanson, Itle, and Edquist, 2024).

Compared to RuFaS v0.8, the updated method results in a **90% decrease** in the manure methane yield.

Take Home: Solid storage emissions are expected to reduce in most cases.

Reason: Addition of heifer specific manure methane potential (PR 2502), removal of bedding VS from empirically estimated solid storage systems (PR 2599), and specification of the effect of mixing in bedded pack systems (PR 2492) resulted in reduced emissions for most solid storage systems.

Table 12: Percent Change (%) in v1.0 carbon intensity predictions from those in v0.8

Difference from v0.8 Results		**Absolute Average**	**Median**	**Min**	**Max**
Total Footprint Change	%	13.88%	-4.65%	-33.10%	31.65%
Footprint Change w/o Feed	%	18.60%	-12.73%	-32.33%	35.51%
Enteric Intensity Change	%	14.56%	17.34%	-1.94%	21.34%
Manure Intensity Change	%	44.25%	-40.11%	-62.86%	62.92%

2.6 Carbon Footprint of Milk - Change from v0.8

- The magnitude and source of differences between v0.8 predictions and v1.0 predictions vary by farm and are affected by the changes in milk and intake predictions as described earlier.
- Comparisons to v0.8 outputs should ignore the feed emissions because the emissions factors were updated after the v0.8 evaluation.

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